Project / System	Elevator System
Prepared By	L Ebert
Version	1.0
Date	Aug 10, 2026
Approved By	O Kipp

Summary

Total Tests	Passed	Failed	Blocked
20	**20**	**0**	**0**

ID	Category	Sub Item	Verification Method
P3-VV-001	CAN/LAN Communication	Implement CAN/LAN communication bridge	Functional test / observation
P3-VV-002	CAN/LAN Communication	Ensure activity in CAN network changes values in database	Functional test / observation
P3-VV-003	CAN/LAN Communication	Changes in GUI change value in database and generate CAN messages	Functional test / observation
P3-VV-004	CAN/LAN Communication	CAN messages result in changes to database and GUI/Indicators	Functional test / observation
P3-VV-005	CAN/LAN Communication	Commands from the LAN network are sent to the controller and move the elevator/indicate floor number	Functional test / observation

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ID	Category	Sub Item	Verification Method
P3-VV-006	Controller Software	Raspberry Pi relays messages between CAN/LAN network	Functional test / observation
P3-VV-007	Finite State Machine	Develop and implement a finite state machine	Functional test / observation
P3-VV-008	Additional Function	Create Maintenance Menu in GUI	Functional test / observation
P3-VV-009	Additional Function	Require Authentication for Maintenance Menu	Functional test / observation
P3-VV-010	Additional Function	Use Database values to change maintenance flags in menu	Functional test / observation

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20	**20**	**0**	**0**

ID	Category	Sub Item	Verification Method
P3-VV-011	Previous Phase Checklist	Car Controller Functionality	Record review / pass-fail confirmation
P3-VV-012	Previous Phase Checklist	Floor Nodes Functionality	Record review / pass-fail confirmation
P3-VV-013	Previous Phase Checklist	Door Open/Close Functionality	Record review / pass-fail confirmation
P3-VV-014	Previous Phase Checklist	Queue Functionality	Record review / pass-fail confirmation
P3-VV-015	Previous Phase Checklist	GUI Complete	Record review / pass-fail confirmation

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ID	Category	Sub Item	Verification Method
P3-VV-016	Previous Phase Checklist	Database Made	Record review / pass-fail confirmation
P3-VV-017	Previous Phase Checklist	Diagnostic Reports Present	Record review / pass-fail confirmation
P3-VV-018	Previous Phase Checklist	Sabbath Mode Functionality	Record review / pass-fail confirmation
P3-VV-019	Previous Phase Checklist	Maintenance Mode (Base)	Record review / pass-fail confirmation
P3-VV-020	Previous Phase Checklist	Floor Announcement Functionality	Record review / pass-fail confirmation

Phase 3 Verification and Validation

Phase 3 expands verification and validation to the CAN/LAN communication bridge, controller message relay b functions, and a previous-phase regression checklist. The following items align with the Phase 3 project requ finalized.

Not Run	Completion %
0	100.0%

Expected Result	Test Procedure	Acceptance Criteria
The CAN/LAN bridge is implemented and can exchange messages between the CAN network, LAN network, database, and GUI as required.	1) Start the CAN/LAN bridge service. 2) Verify bridge connection to the CAN network, LAN network, database, and GUI. 3) Send representative CAN and LAN messages. 4) Confirm messages are	Pass when the bridge starts successfully, connects to all required interfaces, and routes representative messages in both directions without loss or
CAN network activity updates the corresponding database values.	1) Generate or observe a known CAN network event. 2) Inspect the related database record/value. 3) Compare the database value against the expected CAN event. 4) Record evidence.	Pass when each tested CAN event creates or updates the correct database value with no missing or incorrect records.
GUI changes are written to the database and generate the expected CAN messages.	1) Change a supported command/value in the GUI. 2) Inspect the database for the matching update. 3) Monitor the CAN network for the generated message. 4) Compare against the expected CAN	Pass when the GUI change updates the correct database field and produces the correct CAN message within the expected operating time.
Incoming CAN messages update the database and are reflected in the GUI or indicators.	1) Send or observe a known CAN message. 2) Confirm database value changes. 3) Confirm GUI status or indicator changes. 4) Compare all outputs against the expected state.	Pass when the database and GUI/indicators consistently reflect each tested CAN message.
LAN commands are relayed to the controller and result in elevator movement and/or correct floor indication.	1) Send a valid LAN command for elevator movement or floor indication. 2) Monitor command relay to the controller. 3) Observe elevator response and floor indication. 4) Record result and	Pass when valid LAN commands reach the controller and cause the expected elevator movement or floor number indication.

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Phase 3 expands verification and validation to the CAN/LAN communication bridge, controller message relay b
functions, and a previous-phase regression checklist. The following items align with the Phase 3 project requ
finalized.

Not Run	Completion %
0	100.0%

Expected Result	Test Procedure	Acceptance Criteria
The Raspberry Pi relay passes messages between the CAN and LAN networks reliably.	1) Start the Raspberry Pi relay program. 2) Send representative messages from CAN to LAN and LAN to CAN. 3) Observe both network interfaces. 4) Check logs or database records for relay status.	Pass when representative messages are relayed between CAN and LAN without unexpected delay, duplication, or loss.
The finite state machine is implemented and controls elevator states according to defined transitions.	1) Review defined elevator states and transitions. 2) Trigger each key input/event. 3) Observe state transitions and controlled outputs. 4) Compare behaviour against the FSM design.	Pass when the system enters the correct states, performs expected actions, and rejects invalid transitions according to the design.
A maintenance menu is available in the GUI and displays required maintenance options/status values.	1) Open the GUI. 2) Navigate to the maintenance menu. 3) Verify the menu is visible, labelled, and contains required options/status values. 4) Record evidence.	Pass when the maintenance menu is accessible through the GUI and contains the required maintenance controls and indicators.
Access to the maintenance menu requires successful authentication.	1) Attempt to access the maintenance menu without authentication. 2) Attempt access with invalid credentials. 3) Attempt access with valid credentials. 4) Compare access results.	Pass when unauthenticated or invalid users are denied access and valid users are granted access to the maintenance menu.
Database maintenance flag values are reflected accurately in the maintenance menu.	1) Set or update representative maintenance flag values in the database. 2) Refresh or reopen the maintenance menu. 3) Verify menu flags match database values. 4) Change values if	Pass when maintenance menu flags match the database values and update consistently after changes.

Phase 3 Verification and Validation

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functions, and a previous-phase regression checklist. The following items align with the Phase 3 project requ
finalized.

Not Run	Completion %
0	100.0%

Expected Result	Test Procedure	Acceptance Criteria
Car Controller Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Floor Nodes Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Door Open/Close Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Queue Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
GUI Complete has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.

Phase 3 Verification and Validation

Phase 3 expands verification and validation to the CAN/LAN communication bridge, controller message relay b
functions, and a previous-phase regression checklist. The following items align with the Phase 3 project requ
finalized.

Not Run	Completion %
0	100.0%

Expected Result	Test Procedure	Acceptance Criteria
Database Made has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Diagnostic Reports Present has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Sabbath Mode Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Maintenance Mode (Base) has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.
Floor Announcement Functionality has prior verification evidence and remains acceptable for Phase 3 integration.	1) Review prior phase test evidence or demonstration record. 2) Perform a simple spot-check if practical. 3) Record Pass or Fail in this Phase 3 checklist.	Pass when prior evidence exists and no Phase 3 regression is observed. Fail when the item is not functional or evidence is missing.

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iirements and may be customized as implementation details are

Result	Validator / Date	Evidence / Notes
Pass	BG/ 2026-08-17	
Pass	OK/ 2026-08-10	Video recorded
Pass	BG/ 2026-08-14	
Pass	OK/ 2026-08-10	
Pass	BG/ 2026-08-17	

ehaviour, finite state machine implementation, maintenance menu
iirements and may be customized as implementation details are

Result	Validator / Date	Evidence / Notes
Pass	BG/ 2026-08-10	
Pass	BG/ 2026-08-14	
Pass	OK/ 2026-08-16	
Pass	OK/ 2026-08-16	
Pass	OK/ 2026-08-16	

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Result	Validator / Date	Evidence / Notes
Pass	LE/ 2026-08-17	
Pass	LE/ 2026-08-17	
Pass	LE/ 2026-08-17	
Pass	BG/ 2026-08-17	
Pass	LE/ 2026-08-17	

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Result	Validator / Date	Evidence / Notes
Pass	LE/ 2026-08-17	
Pass	LE/ 2026-08-17	
Pass	BG/ 2026-08-17	
Pass	BG/ 2026-08-17	
Pass	LE/ 2026-08-17	